K SHREEKRISHNA UPADHYAYA

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PROFILE

Software engineer who writes low-level C — memory allocators, threading, virtual memory, language runtimes — and also works through specification-driven, AI-assisted workflows on separate projects.

EDUCATION

VIVEKANANDA COLLEGE OF ENGINEERING AND TECHNOLOGY, PUTTUR DK, Karnataka
Bachelor of Engineering in Artificial Intelligence & Machine Learning September 2023 – Present
Cumulative GPA (till 6th semester): 8.69/10.0

PROJECTS

Sn Project | C, x86_64 Assembly December 2025 – Present
Collection of 13 cross-platform C libraries providing reusable building blocks for native software development, extracted from an experimental game engine and subsequent projects as the repeated-need pattern emerged.

- Designed 7 memory allocator strategies (linear, stack, pool, freelist, queue, ring buffer, frame) behind a unified `SnMemoryAllocator` interface; backed allocators with a two-phase virtual memory subsystem using `mmap/VirtualAlloc` reserve-commit pattern
- Built cross-platform threading primitives (mutex, rwlock, semaphore, spinlock, condvar, thread) using opaque ABI-stable byte buffers that never leak platform types to headers; hand-wrote atomics in x86_64 inline assembly with explicit `lfence/sfence/mfence` placement for memory ordering semantics
- Architected a two-tier logging system: ring buffer for low-latency fast-path enqueue with heap-backed overflow fallback; thread-safety is optional, left to user-provided lock hooks; pluggable sinks via function-pointer abstraction
- Applied variable-length quantity encoding for alignment bookkeeping, reused independently across two unrelated allocators
- Extracted testing infrastructure from Snuk into SnTest, a reusable unit testing framework maintained as part of the ecosystem

Snuk | C April 2026 – Present
Embeddable scripting language with expression-oriented semantics and a custom lightweight runtime (6K lines of C).

- Implemented reference-counted memory management with explicit strong/weak pointer separation to eliminate runtime memory leaks
- Built a lexical analyzer, recursive-descent parser, and tree-walking interpreter with first-class functions and closures
- Maintained cross-platform build and test infrastructure via GitHub Actions CI/CD

DADS (Detection and Analysis of Dysfluencies in Speech) | Python, PyTorch, Librosa September 2025 – November 2025
Stuttering dysfluency detection system on the SEP-28k dataset, exploring both per-type and multi-label CNN architectures.

- Trained 5 binary CNN classifiers (one per dysfluency type: prolongation, block, sound repetition, word repetition, interjection) on mel-spectrogram features with per-type hyperparameter search across window size, hop length, and mel bands
- Explored a CNN-LSTM multi-label classifier as an alternative to separate per-type models
- Built a PyQt5 desktop application with real-time audio recording, playback, spectrogram visualization, and background stutter detection inference

The Major Project extends this work with a Wav2Vec2 + CNN-transformer hybrid architecture.

FinDiary | Go, PostgreSQL, Flutter, gRPC July 2026 – Present
Ongoing personal finance tracking application developed through Specification-Driven Development, used deliberately to explore agentic AI workflows as a counterpoint to the hand-written systems work elsewhere in this resume — specifications are authored and refined before AI generates implementation code, with output reviewed for architectural consistency.

- Architecting Go backend services with PostgreSQL persistence and gRPC-based client-server communication
- Directing development of a Flutter cross-platform client, with offline-first local state management planned

hash shell | C January 2026 – March 2026
Open-source contributor with 17 merged pull requests to an active codebase.

- Proposed and led a multi-PR refactoring initiative (8+ PRs) extracting helper functions and aligning the codebase with upstream coding standards
- Fixed tab-completion prefix errors and multi-line continuation bugs in quoted strings

Additional Projects:
Regex Engine (Thompson NFA construction and simulation in C), StateFlow (DFA/NFA simulator).

SKILLS

Languages & Build: C, C++, CMake, Python, x86_64 Assembly (inline/MASM), Go, Java, POSIX Shell
Systems: POSIX APIs, Memory Management, Virtual Memory, Threading & Synchronization, File Systems, Networking, Dynamic Loading
Development: Git, Linux, Typst, Vim, CI/CD (GitHub Actions), gRPC, REST APIs, PostgreSQL, Flutter
Achievements: Winner of Nexathon 2025 (SDIT) and DevHack 2025 (Sahyadri College)